

Power Series & Their Properties

MATH 146 – Calculus II

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Taylor Polynomials & Infinite Series

Recall: Taylor Polynomials:

Definition:
$$p_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x - a)^k.$$

Taylor's Theorem:
$$|f(x) - p_n(x)| = |R_n(x)| \leq \frac{M}{(n+1)!} (x - a)^{n+1}.$$

Recall: Infinite Series $\sum_{k=1}^{\infty} a_k$:

$\sum a_n$ may or may not converge, we have many “tests” to determine if a series converges.

Power Series

Power Series

A **power series** is an expression of the form $\sum_{n=0}^{\infty} c_n x^n$ (or more generally $\sum_{n=0}^{\infty} c_n (x - a)^n$, where a is the **center**).

A power series **converges** at x is the numerical series $\sum_{n=0}^{\infty} c_n x^n$ converges.

A power series $\sum_{n=0}^{\infty} c_n (x - a)^n$ defines a function for each x at which it converges.

Examples: Power Series

Example: $\sum_{n=0}^{\infty} \frac{1}{n!} x^n.$

Example: $\sum_{n=0}^{\infty} x_n.$

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Radius and Interval of Convergence

Radius of Convergence

For each power series $\sum c_n(x - a)^n$, there is a “number” R such that $0 \leq R \leq \infty$ and the series converges for $|x - a| < R$ and diverges for $|x - a| > R$.

Example: What is the *radius of convergence* of $\sum_{n=1}^{\infty} \frac{1}{n} x^n$? What is the *interval of convergence*?

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Integration and Differentiation of Power Series

Polynomials are the best functions imaginable for calculus, and power series are essentially just infinitely long polynomials with all the benefits.

Theorem

If $f(x) = \sum a_n x^n$ has radius of convergence R , then:

1. $f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}.$

2. $\int f(x) dx = \sum_{n=1}^{\infty} \frac{a_n}{n+1} x^{n+1} + C$

and both of these have the same radius of convergence.

Examples: Power Series

Example: Let $f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} x^n$. What is $f'(x)$? What is $\int f(x) dx$?

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Functions and Power Series

Some well-known functions can be given by power series expansions instead of their usual formula – and computations can be carried out using the power series expression.

Example (Geometric Series):

$$\frac{1}{1-x}$$

Example (“The Substitution Trick”):

$$\frac{1}{1+3x^2}$$

Examples: Power Series

Example: Find a power series representation for $\ln(1 - x)$ centered at $x = 0$.

Example: Find a power series representation for $\ln\left(\frac{1+x}{1-x}\right)$ centered at $x = 0$.

Examples: Power Series

Example: Find $\int_0^{\frac{1}{\sqrt{3}}} \frac{1}{1+x^2} dx$.